

Shortleaf pine cover loss near Stone and Taney Counties Missouri using Landsat 4-5 (TM Sensor) satellite imagery manipulation with supervised classification

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Abstract

Prior to European settlement on what is now the United States of America, nearly half the country was forested. As of 2010, 751,255,00 acres of forested land remains, a number that represents one-third of the country, (Becker, n.d.). Approximately 32% (13.9 million acres) of the state of Missouri is forested providing major habitat for wildlife, botanical diversity, soil and water quality protection, as well as temperature and native plant protection. In the very southwestern portion of Missouri lies a section of the shortleaf pine (*Pinus echinata*) range, a species with high rates of deforestation across the eastern half of the country. This study utilizes historic (1984) and modern (2011) multitemporal Landsat 4-5 TM satellite imagery to analyze a section of Missouri in Stone and Taney counties to show the land cover change from urbanization contributing to forest degradation within the shortleaf pine range. Composite images are created with using appropriate RGB band composites in *Clark Labs TerrSet* and then classified for comparison using *ESRI ArcMap* software. By showing the land cover change across this area over time, the imagery can present conifer forest health degradation.

Keywords: Degradation, Landsat, Supervised Classification

1. Introduction and Study Area

1.1 Introduction

Missouri is the 18th most populous state in America and in its southwestern corner lies a small portion of the shortleaf pine range nestled in the Ozarks that extend into Arkansas. Not representing a metropolis, the town of Branson and surrounding rural towns hold a sample site that encompasses increasing human population along with built environment. The use of land cover change analyzation is useful for everything from soy crop depletion to redwood density change. According to (NOAA), remote sensing is the science of obtaining information about objects or areas from a distance, typically from aircraft or satellites. Taking this data,

manipulating and presenting in the proper software allows cover models for analysis over specified time periods. This study is meant to provide a particular example of expanding built environment and tree density depletion. The shortleaf pine has a wide, but dwindling range in southeastern U.S. This section against the Ozarks presents a particularly challenging problem as tourism and urbanization takes place. Shortleaf pine generally competes best with hardwoods or other pine on dry, shallow, infertile, and rocky soils, although it achieves its best growth on more fertile, deeper, soils, (Scott, n.d.). The pine grows particularly well against the guarded bluffs that makeup much of the Ozark territory in Arkansas and Missouri. As population push into these areas, degradation of not only native tree species begins but previously vacant soils and grasslands. This area is chosen to display five main land cover types against the shortleaf growing range amongst the Ozark mountains.

1.2 Study Area

Branson overlaps the border of Taney and Stone counties in Missouri and with the Arkansas state border just to its south. While vegetation in the area is in no immediate danger of complete loss, this study zone sits on the edge of the coniferous shortleaf pines current territorial range. This area serves as a great reference that includes budding urban traffic and creation with specific coniferous species loss amongst a mostly rural micropolitan area in the Ozarks. Figures 1 and 2 show the specific sample area.

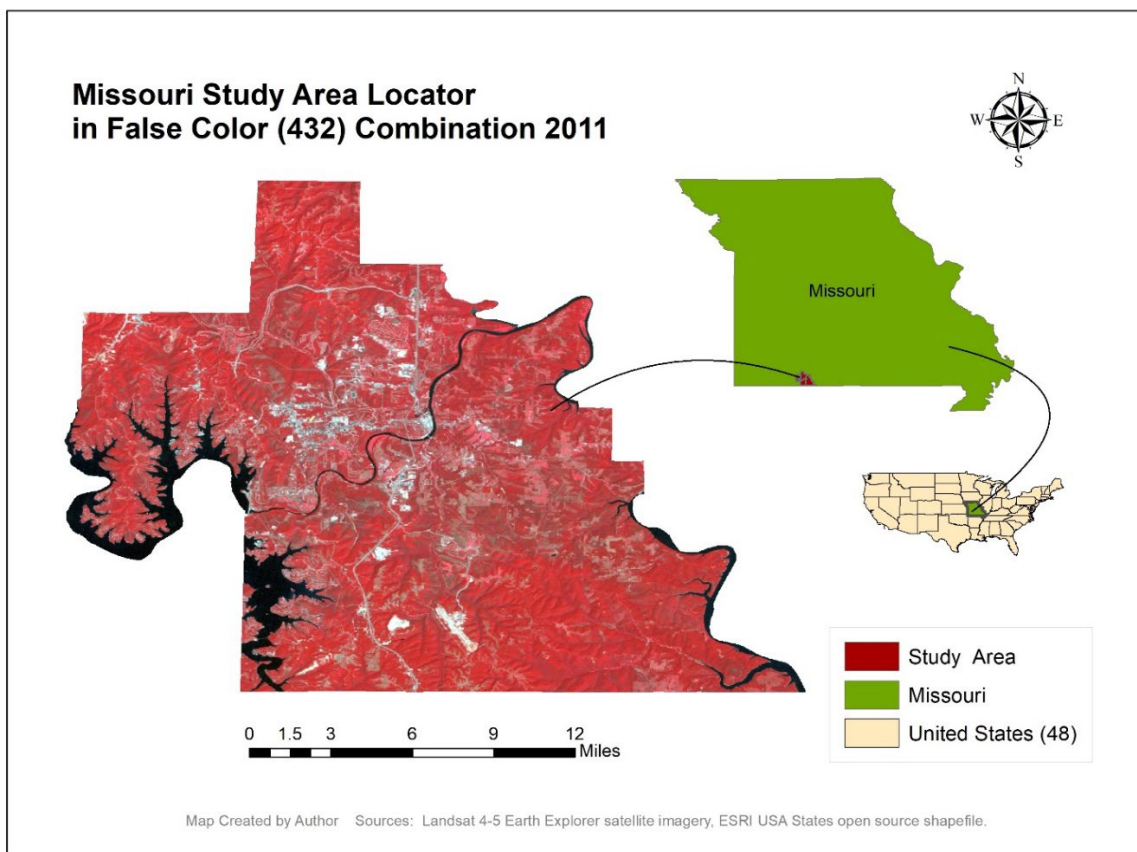


Figure 1. Location of Study Area

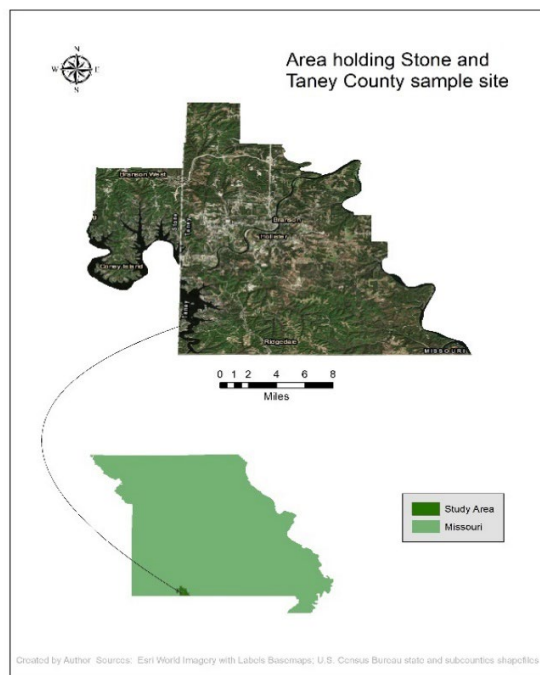


Figure 2. Locator Map 2

2. Data and Methodology

2.1 Data

Landsat 4-5 Thematic Mapper multi-temporal satellite data is used for the study. It holds seven spectral bands of 30 m spatial resolution (aside from thermal band 6 that holds a 120 m spatial resolution). Table 1 displays the specificities of Landsat TM 4-5 data used.

Table 1. Landsat TM 45 spatial characteristics

Bands	Wavelength μm (micrometers)	Resolution m (meters)
Band 1 - Blue	0.45-0.52	30
Band 2 - Green	0.52-0.60	30
Band 3 - Red	0.63-0.69	30
Band 4 - Near Infrared	0.76-0.90	30
Band 5 - Shortwave Infrared 1	1.55-1.75	30
Band 6 - Thermal	10.40-12.50	120* (30)
Band 7 - Shortwave Infrared 2	2.08-2.35	30

Source: United States Geological Survey EarthExplorer

Table 1 reflects the characteristics for the data acquired on both 18 September 1984 and 28 August 2011.

2.2 Methodology

This study is meant as a presentation and valuation of satellite imagery analyzation in TerrSet 1.0 and ArcMap 10.5 software for the purpose of measuring the effects on forest health due to urbanization and built human environments. TerrSet is an integrated geospatial software system for monitoring and modeling the earth system for sustainable development. The TerrSet system incorporates the IDRISI GIS Analysis and IDRISI Image Processing tools along with a constellation of vertical applications (Clark Labs, n.d.). False color infrared composite combination band 4-3-2 (NIR, red, green) is created and processed digitally using TerrSet's image composite utility on both images (1984 and 2011). These images are then clipped to the

study area using the GIS tools in ArcMap. From here the final false color study area images are brought back into TerrSet and classified through supervised digitization following specific pixels to identify five image classes. False color composites work in this case as the images are useful in vegetation identification and species health. In this case, the health of deciduous trees is measured where high density tree growth is displayed in brighter red versus sparse growth showing as darker red pixels. This digitization is used to create polygons for five separate classes (water, healthy trees, lower health trees, grassland/soil, built environment) with unique (1-5) identifiers for the land cover type. After identifying necessary training sites, spectral signatures are created that analyze the reflected energy from the land and water sources within the images. After the signature development process, images are run through a maximum likelihood classifier with equal probabilities for each signature/class for an accurate output. Figure 2 shows maximum distance classification results examples for the five spectral signatures.

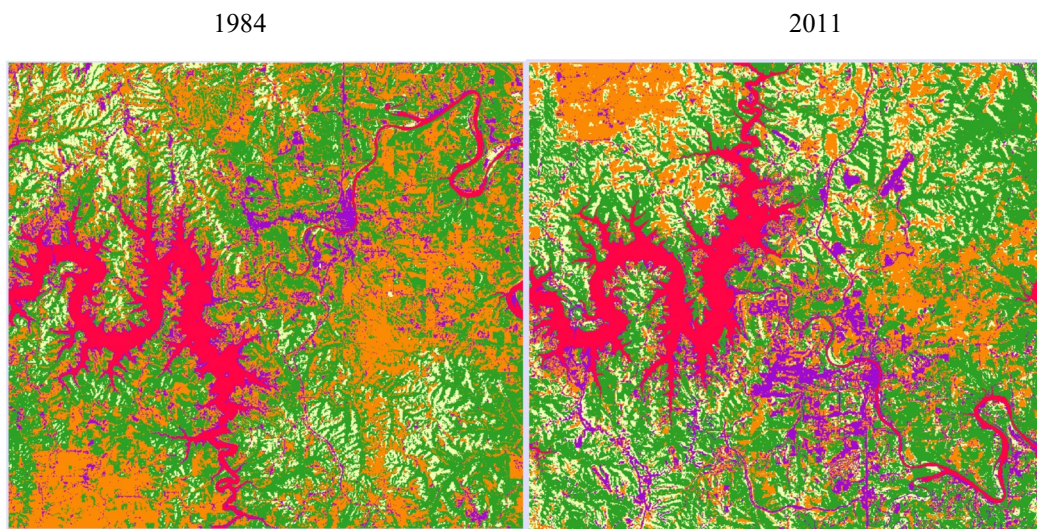


Figure 2. Example Maximum Distance output for study area and extent

The 1984 and 2011 maximum distance classifiers are then crossed through TerrSet's Land Cover Modeler to visually display area change maps and area change tables. Final change maps presenting the rise of unhealthy forest (less dense) containing the shortleaf pine and built

environment increase are then complete for comparison. The methodology process is presented in Figure 3.

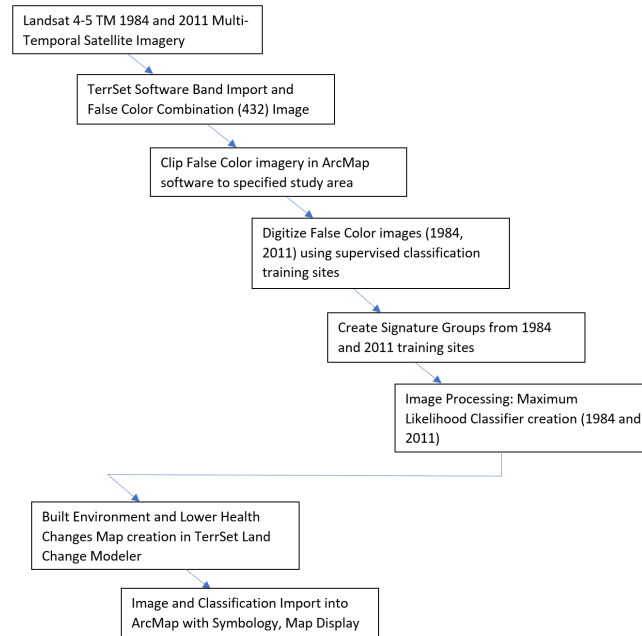


Figure 3. Study Methodology and Process

3. Results and Discussion

The satellite images used for this study were broken down into a sub-counties shapefile from the U.S. Census Bureau Tiger/Files database. By separating the classes needed from an image, the IDRISI image processing task within TerrSet will calculate area based on pixels and other specific area measure (in this case acreage). The passively gathered Landsat 4-5 TM data is taken from peak growing months for all tree species. Thorough examination of former and current landcover was completed. By then analyzing pixels and cell counts, the scenes were broken into five manageable classes.

3.1 Classification maps and statistics

3.2 Study area accuracy assessment

With supervised classification, it is necessary to measure accuracy alongside the change being studied.

Class	Reference Data	
	Lower Health Conifer	Built Environment
Lower Health Conifer	125	67
Built Environment	87	139
Totals	212	206
Producer's Accuracy		
Lower Health Conifer	125/212	59%
Built Environment	139/206	67%

Table 2. Producer's Accuracy Assessment

Land Cover Type	1984			2011			% Change Area (acres) (1984 - 2011)
	Area (Acres)	% of Area	Pixels	Area (Acres)	% of Area	Pixels	
Water	16154.55319	7.22%	72638	18330.0514	8.20%	82420	0.97%
Healthy Conifer	19363.09058	8.66%	87065	35784.07669	16.00%	160901	7.34%
Lower Health Conifer	90663.92144	40.55%	407665	91444.98356	40.90%	411177	0.35%
Grassland/Soil	76948.40821	34.41%	345994	51923.06201	23.22%	233469	-11.19%
Built Environment	20472.4123	9.16%	92053	26120.21205	11.68%	117448	2.53%
Total	223602.3857	100.00%	1005415	223602.3857	100.00%	1005415	

Table 3. Maximum likelihood classification area change (1984-2011)

3.3 Further analysis

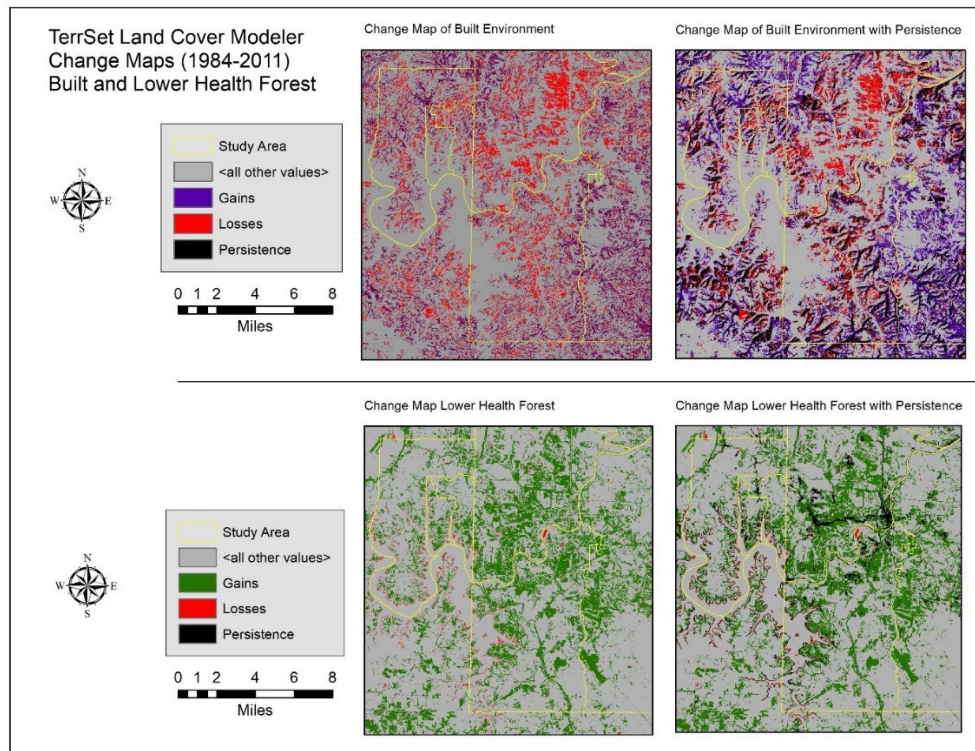
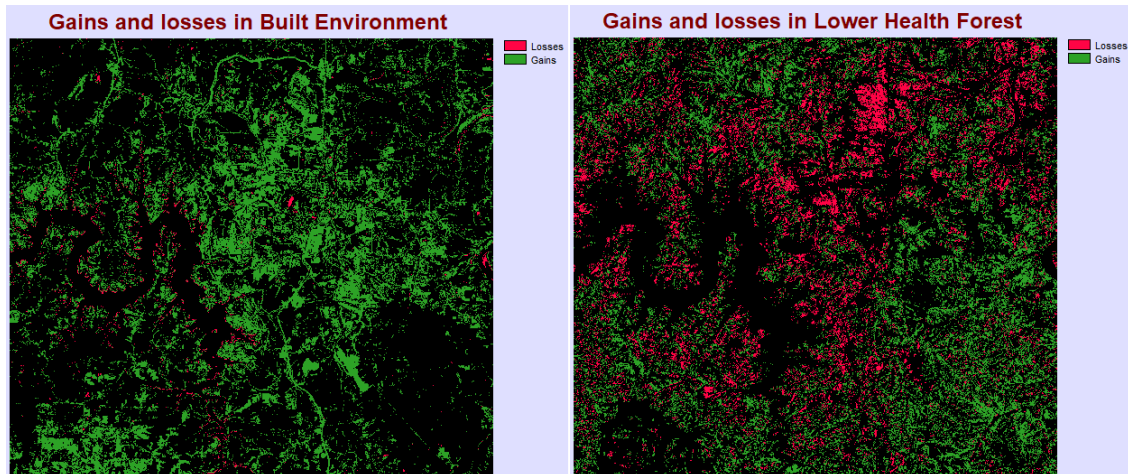


Figure 5. Change Maps of built and lower health conifer forest gains/losses

4. Conclusion

Satellite image digitization and classifiers effectively measure land cover change. By retrieving Landsat imagery and combining appropriate multispectral bands, Earth's land cover and resources can successfully be viewed, monitored, and analyzed for a multitude of measures.

Specifically, by creating a false-color color infrared composite image combining the Near Infrared (NIR) (4), Red (3), and Green (2) bands and using a select area, this color and image manipulation is valuable in exploring forest cover health and expanding urban sprawl. By using the near infrared band, enhanced spectral separation improves user identification. As the spectral reflectance for vegetation is based on chlorophyll and water absorption in leaves, varying forest cover health can be deciphered. Healthier leaves better absorb electromagnetic energy which show as brighter reds in the combination image, with trees with less healthy leaves shown in darker reds.

This study is meant as an illustration on the effects of urban and population advance on forest health. While the necessity for protection may lie elsewhere, the Stone and Taney County Missouri sample is offered as it displays urbanization amongst a micropolitan area and spreads along the Ozarks and within the shortleaf pine range. The sample was meant to show the damage urbanization may cause to native conifers and reason for regeneration. With initiatives currently working to maintain and regenerate the shortleaf pine, investigation should continue to create proper forest management as built human environments breach edges of this yellow pine range in the very southwestern part of Missouri.

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